AASTHA RAWAT

Full-Stack & AI Systems Builder | Rapid MVP Prototyping Engineer

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PROFESSIONAL SUMMARY

High-velocity Full-Stack and AI Engineer specializing in edge computer vision, optical signal processing, and rapid MVP engineering for tech startups. Creator of **BioTwin Labs** (optical vital scanner), **SmartAttend+** (college verification infrastructure), and **DevPulse AI** (code security audit engine). National sports medalist (Khelo India Rugby) and decorated NCC Cadet with exceptional execution speed and disciplined leadership.

TECHNICAL TOOLKIT

Languages & Web: Python, TypeScript, JavaScript, C++, React, Vite, Node.js, HTML5/CSS3

AI & Computer Vision: Optical PPG (Photoplethysmography), Signal Processing (DSP), OpenCV, Autonomous Agents

Mobile & Backend: Capacitor 7 (Android SDK 35 Native), FastAPI, Flask, REST APIs, WebSockets, PWA

Cloud & Infrastructure: Firebase Auth & Firestore, GitHub Actions CI/CD, Edge Deployment, Google Play Console

FLAGSHIP PROJECTS & MVP PORTFOLIO

BioTwin Labs — Edge-AI Health Diagnostics & Digital Twin | *Optical PPG, DSP, Android SDK 35, Python*

- Engineered India's first contactless optical vital scanner converting smartphone camera sensors into clinical photoplethysmography (PPG) monitors measuring Heart Rate (BPM), SpO2, and HRV at 30-60 FPS without wearables.
- Developed an acoustic vocal biomarker engine for non-invasive respiratory indicators and a 24/7 AI clinical reasoning copilot.
- Shipped production native Android App Bundle (.aab) targeting SDK 35 and 24/7 Global Edge deployment: https://aastharawat47849-oss.github.io/biotwin-labs/

SmartAttend+ — AI-Powered College Attendance & Verification Infrastructure | *React, TypeScript, Firebase, OCR*

- Architected an automated event attendance and multi-level approval system for engineering colleges with real-time broadcast to subject teachers.
- Built an integrated AI Application Writer, OCR document scanner, and co-signed verifiable digital certificates hub.
- Praised and requested for live demonstration by engineering department faculty at OIST Bhopal: https://aastharawat47849-oss.github.io/smartattendance/

DevPulse AI — Autonomous Security & Code Auditing Agent | *Python, FastAPI, Security AST, REST API*

- Created an autonomous DevSecOps AI agent that scans repositories and AST trees to flag hardcoded secrets, SQL injection vectors, and anti-patterns before production deployment.
- Open-source codebase: https://github.com/aastharawat47849-oss/DevPulse-AI

FPS Ghost Hunt — 3D WebGL Interactive Game | *WebGL, 3D Physics, JavaScript Game Loop*

- Built a 3D First-Person browser game implementing real-time spatial audio, lighting physics, and low-latency frame loop optimization.

EDUCATION

Bachelor of Technology (B.Tech) — Electronics & Communication Engineering (ECE)
Oriental Institute of Science & Technology (OIST), Bhopal | 2024 – 2028

LEADERSHIP & NATIONAL HONORS

- **Gold Medal & Trophy:** National Cadet Corps (NCC) AITC / ATC Camp — awarded for exemplary drill, obstacle navigation, and unit leadership.
- **National Silver Medalist (2nd Position):** Khelo India Rugby Championship — national-level competitor with proven tactical discipline.
- **Nodal Gold & State Bronze Medal:** State Aquatics / Swimming Championship — competitive speed champion.
- **All India Trekking Expedition (AITE):** Amarkantak Trek II (2025) — completed national high-altitude survival expedition.